

Writing Plain Language Conclusions

This worksheet is meant to help you obtain an intuition of what counts as “statistical jargon” and how to re-explain concepts to a non-statistical audience. In particular, it is important for you to be able to interpret results in plain language.

Where is the Statistical Jargon?

The key question to ask yourself is, “if you read this report in high school, would you be able to understand what is going on?”

(Previous years when I gave this advice, some students claimed they had no brain in high school, so it was bad advice. I hope this cohort of STA304 students could confidently claim that their high school self would be able to briefly understand concepts on a high-level!)

Below is a list of sentences to interpret the results of survey data. Determine below if these sentences contain statistical jargon. If it does, please re-write it so a non-statistical audience can understand.

(This was a past tutorial activity. I just provided the answers in this version below.)

1. Through thorough statistical testing, we can confidently assert that 80% of students have a clear preference for public transport when compared to other choices.

ANSWER:

NO. This sentence is perfectly acceptable!

2. The 95% confidence interval for the mean of Fall 2023 STA304 students' cGPA is (3.0, 3.3).

ANSWER: (if you know of other sentences, please suggest them to students.)

YES. We can instead write something like, “we are 95% certain that the mean of Fall STA304 students’ cGPA is between 3.0 and 3.3.”

3. Within our linear regression model $Y = \beta_0 + \beta_1 X + \varepsilon$, where Y represents the predicted students’ cGPA and X represents the number of lectures students attended, we determined that the coefficient of determination (R-squared) is 0.85.

ANSWER:

YES. 85% of the variations in students' CGPA can be explained by the number of lectures they attended.

4. The null hypothesis was rejected at the 0.05 significance level.

ANSWER:

YES. We found strong statistical evidence that our proposed hypothesis is true.

5. N represents the total number of Fall 2022 STA304 students, and B represents the bound of error, which informs the audience the extent to which your estimate might differ." I.e., if you obtain a proportion of 0.5 with a bound of 0.1, then your estimated proportion could be between (0.45, 0.55) instead.)

ANSWER:

NO. This sentence is acceptable! In particular, clearing up concepts in layman's terms is good if you must elaborate on important details.

6. We can confidently claim (with 95% certainty) that more than 50% of STA304 students do not believe that hotdogs are sandwiches.

ANSWER:

NO.

7. Our statistical model reveals a negative relationship between cannabis usage and lecture attendance. Another way to interpret the validity of our model is to analyze R^2 , the coefficient of determination, which tells us how accurately lecture attendance can be linearly modeled through cannabis usage.

ANSWER:

This is a bit borderline but I am heading towards YES. Although they explain R^2 , the issue is "linearly modeled" and may be a bit hard for others to interpret. Similarly with the "negative relationship" line.

Here's an alternative:

Our statistical analysis shows that when people use cannabis more, they tend to attend fewer lectures. To gauge how well our model predicts lecture attendance based on cannabis usage, we can look at R^2 (the coefficient of determination), which helps us understand how closely the two are linked if modeled by a straight-line"